enabling the identification of opportunities to remove bottlenecks and improve throughput. Digital twins also see extensive use in the training of personnel in these industries to safely operate in hazardous environments.

AUTOMOTIVE

The automotive industry has been an enthusiastic adopter of digital twins across many parts of the industry for many years. Digital twins are used in the product development and engineering phase to simulate aerodynamic performance, reducing the need to build physical prototypes for testing. As the transition to electric vehicles gathers pace, with highly instrumented and connected vehicles with complex onboard software, digital twins are increasingly being used to maintain a digital representation of the state of each vehicle, as well as changing its state through software updates.

AGRICULTURE

As the United Nations predicts the global population to peak at around 10.3 billion people around 2080, up from approximately 8 billion people at the time of writing, agriculture and food production will need to become more efficient, improving yields and reducing waste. There are many ways in which complex agricultural processes can benefit from the living digital representation that a digital twin provides. Accurate and up to date data about plant, animal, and machine health allows farmers to more efficiently allocate resources. Digital twins of global weather conditions, such as the Destination Earth digital twin being developed by the ECMWF, contribute to more accurate forecasting of weather events that are so important in food production.

NOTE

The United Nations provides extensive datasets related to global population here https://population.un.org/wpp/.